

Event Density — Predictions and Falsifiers

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Event Density (ED) is a discrete, relational substrate whose one special feature is that becoming is an *irreversible commitment* — the arrow of time written into the law rather than the boundary conditions. From that single inversion the program derives an emergent Einstein weak field, a specific gravitational class (khronometric), the galactic modified-dynamics (MOND) phenomenology, and the cosmological-constant scale — one field, one scale, several roles.

This page exists because a theory worth taking seriously must be able to lose. Below is every place the program exposes itself to data, sorted by how decisive the test is. Each entry states the claim, the number, how to check it, and **what specific observation would kill it**. Nothing here is asserted as confirmed physics; the program is conditional on its postulates and is offered as a set of falsifiable claims, not a finished theory. The point is to make the failure modes explicit, so the work can be checked rather than merely believed.

Sources are the gravity-line papers (GR-I ... GR-IV, KM-I, KM-II, the One-Field letter, and the numbered substructure 025–038.6), archived with DOIs.

Tier 1 — Postdictions already on the table (reproduced, not tuned)

These are known facts the substrate reproduces without free parameters. They are not new predictions, but they are real constraints the framework had to clear.

Claim	Value	What would have killed it
Factor-of-two light bending	deflection exactly twice the Newtonian value; ray-traced ratio 2.09 vs 2.000, conformal control at 0	a sub-Einstein (half-strength) or zero bending — i.e. a scalar-gravity metric (GR-I)
Gravitational-wave speed	$c_T = c$ as a structural identity (one causal cone)	GW170817 already required $ c_T/c - 1 \lesssim 10^{-15}$; a generic theory of this class would have failed it (GR-II)
Newtonian limit & Newton's G	$\nabla^2 b \sim \rho$; $G = c^3 \ell_P^2 / \hbar$ (form derived, value inherited through one scale)	failure to recover Poisson gravity, or a second independent constant (Paper 027)

Claim	Value	What would have killed it
The Milgrom coincidence	the galactic transition scale comes out as $a_0 = cH_0/(2\pi) \approx 1.1 \times 10^{-10} \text{ m/s}^2$, a parameter-free $\sim 10\%$ match	a transition scale unrelated to cH_0 (Paper 029)

Tier 2 — Live observational tests (current and near-future data can decide)

These are the fronts where ongoing measurements can confirm or kill the program *now*.

- **Preferred-frame parameters** α_1, α_2 (*the sharpest front; GR-IV*). The theory predicts these are *unobservably small* — suppressed below the experimental bounds ($|\alpha_1| < 2.5 \times 10^{-5}$, $|\alpha_2| < 10^{-7}$) by tens of orders of magnitude, because the preferred-frame coupling scales with the local density of committed events relative to the Planck density. **Kill condition:** a measured α_1 or α_2 at or above its bound in any sub-Planck-density environment (lunar laser ranging, pulsar timing, solar-spin alignment).
 - **Wide binary stars near a_0** (*KM-I, One-Field*). In the unscreened modified-dynamics regime, wide binaries with internal accelerations near a_0 should depart from Newtonian dynamics. **Kill condition:** a clean Newtonian result at the relevant separations and precision kills the unscreened galactic sector — and ED has no vector field to repair it with.
 - **The baryonic Tully–Fisher relation** (*Paper 031*). Predicted slope **exactly 4** with **zero intrinsic scatter** in the deep regime, $v_{\text{flat}}^4 = GM_b a_0$. Current data: slope 3.95 ± 0.08 , ~ 0.1 dex scatter. **Kill condition:** a slope significantly different from 4, or irreducible intrinsic scatter, beyond observational uncertainty.
 - **The lensing identity** (*KM-I*). Galactic weak lensing should track the modified-dynamics potential with no gravitational slip beyond $O(\Phi)$ — no extra lensing mass relative to dynamics. **Kill condition:** a measured slip, or a lensing-vs-dynamics mismatch the metric-only coupling cannot accommodate.
 - **Activity-dependent weak lensing** (*Paper 038.6, a genuinely new prediction*). Dynamically active systems (star-forming, AGN, unrelaxed) should show *enhanced* lensing relative to quiescent systems of the same baryonic content, via a mobility-gradient correction. **Kill condition:** no activity dependence once an observable separating it from ordinary baryonic-feedback effects is identified (the discriminating observable is itself an open item).
 - **A scalar gravitational-wave polarization** (*GR-II/GR-IV*). The khronon is a physical scalar mode at light speed in vacuum; in principle it is a polarization distinguishable from General Relativity’s two tensor modes. **Caveat (GR-IV):** because the khronon decouples in low-density regions, this signature is faint and near-matter, not a clean vacuum signal — a target for the long term, not a present test.
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Tier 3 — Open numbers the program must hit (computations with fixed targets)

These are not yet computed from first principles, but the *target is already pinned* — so the future computation either lands on the number or breaks the claim. “A theory with a number to miss is a theory that can lose.”

- **The cosmological-constant coefficient** (*KM-II, One-Field, Paper 038.5*). The one- Λ thesis requires the ab-initio substrate boundary integral to reproduce the effective-theory constant $\mathcal{W}_0 = -24\pi^2 \Omega_\Lambda \approx -162$. **Kill condition:** the substrate computation yields the wrong sign or a magnitude off by more than $O(1)$.
 - **The cluster / CMB dust sector** (*KM-II*). The cosmological sector must deliver dust-like clustering (sound speed ≈ 0) of the right abundance at recombination, through a single regulator dial constrained by five independent filters. **Kill condition:** no dial member satisfies all five filters simultaneously, or the CMB acoustic peaks cannot be fit.
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Tier 4 — Structural kill conditions (where the theory’s own logic exposes it)

- **The quantum/relativity link** (*GR-IV*). The preferred-frame safety follows from commitment being *sparse* — and sparse commitment is the *same* structural fact that gives ED quantum mechanics (dense commitment is the quantum Zeno limit, which would freeze all evolution). **Kill condition:** an experiment that forces commitment to be dense — continuous, every-tick determination — would simultaneously break the α_1 suppression and ED’s quantum sector. They stand or fall together.
 - **The single causal cone.** All matter and gravity share one cone, so Lorentz violation is universal, not species-dependent. **Kill condition:** a measured *differential* Lorentz violation between matter species (the 10^{-20} -level bounds), which the single-substrate architecture forbids.
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The honest standing

None of this is confirmed. The program inherits Newton’s constant, the Planck scale, and H_0 from observation; it rests on thirteen postulated primitives and a small number of labeled load-bearing assumptions; and several of the numbers above (the cluster sector, the ab-initio Λ integral) are open computations. What the program offers is not a proof but a posture: a discrete, irreversible substrate that reproduces the architecture of gravity, concentrates its departures from Einstein at a single locus (the arrow of time), and — at every place it touches data — says exactly what would prove it wrong.

If any Tier-1 result had failed, the program would already be dead. The Tier-2 fronts are where present data can kill it. The Tier-3 numbers are where its own future computations can. That is the whole claim: it specifies precisely where to look to find out.

A standing falsification map for the Event-Density gravity program. Postdictions (Tier 1) cleared; live observational tests (Tier 2: α_1, α_2 , wide binaries, BTFR slope-4, the lensing identity,

activity-dependent lensing); pinned open numbers (Tier 3: $\mathcal{W}_0 = -24\pi^2\Omega_\Lambda$, the cluster/CMB dust sector); structural kill conditions (Tier 4: the quantum-relativity link, the single cone). Conditional on the program's postulates; nothing asserted as confirmed. Full derivations and audits in the gravity-line papers (GR-I ... GR-IV, KM-I/II, One-Field, 025-038.6).